

HEINOLA ASEMANAUS, Finland

WMO: 027680

Lat: 61.200N

Lon: 26.049E

Elev: 302

StdP: 14.54

Time Zone: 2.00 (EUE)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -11.0	(c) -5.2	(d) -16.1	(e) 2.3	(f) -10.8	(g) -9.5	(h) 3.3	(i) -4.9	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.8	82.6	64.8	79.0	63.1	75.5	61.6	68.4	77.2	66.1	74.8	63.9	72.1	N/A	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.2	94.1	71.9	62.9	86.9	69.6	60.8	80.4	67.0	32.8	77.9	31.0	75.1	29.3	72.0	74.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) N/A	(b) N/A	(c) N/A	(e) -12.4	(f) 85.1	(g) 8.7	(h) 4.5	(i) -18.7	(j) 88.4	(k) -23.7	(l) 91.0	(m) -28.6	(n) 93.6	(o) -34.9	(p) 96.9		
(4) N/A	(5) N/A	(6) N/A	(e) -12.8	(f) 69.5	(g) 8.6	(h) 3.3	(i) -19.0	(j) 71.8	(k) -24.0	(l) 73.8	(m) -28.9	(n) 75.6	(o) -35.1	(p) 78.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	41.6	18.4	22.6	28.2	39.2	51.7	58.6	64.1	60.8	52.1	40.6	34.3	27.1
	DBStd	17.05	12.20	11.06	8.23	6.46	7.75	6.37	6.26	5.42	5.90	6.29	6.74	9.73
	HDD50	4360	981	768	676	331	73	8	0	1	47	295	470	710
	HDD65	8660	1446	1188	1141	775	416	210	95	152	387	755	920	1175
	CDD50	1285	0	0	0	6	125	265	437	336	111	5	0	0
	CDD65	110	0	0	0	0	3	18	67	22	0	0	0	0
	CDH74	1068	0	0	0	0	85	208	596	179	0	0	0	0
	CDH80	230	0	0	0	0	9	32	149	40	0	0	0	0
	WSAvg	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	PrecAvg	24.0	1.7	1.4	1.3	1.2	1.7	2.5	2.8	2.6	2.1	2.4	2.2	1.9
(15) Precipitation	PrecMax	31.0	3.4	3.2	2.3	2.4	4.3	3.9	5.4	4.7	5.2	5.9	5.2	3.7
	PrecMin	19.8	0.3	0.1	0.1	0.4	0.4	0.7	0.7	0.4	0.8	0.4	0.2	0.4
	PrecStd	2.9	0.7	0.8	0.6	0.6	1.0	0.9	1.4	1.2	1.1	1.3	1.0	0.7
	PrecStd	2.9	0.7	0.8	0.6	0.6	1.0	0.9	1.4	1.2	1.1	1.3	1.0	0.7
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	38.6	41.8	49.9	67.9	80.7	83.8	88.1	84.9	70.8	57.0	50.7	47.0
		MCWB	37.1	36.6	39.9	50.8	58.3	63.6	67.6	66.2	60.9	52.5	48.3	45.7
	2%	DB	36.7	38.1	44.8	61.1	76.1	79.7	84.1	78.3	66.8	53.8	47.9	41.4
		MCWB	35.8	35.7	37.4	46.5	57.8	61.9	66.4	63.7	58.6	50.7	46.6	40.0
	5%	DB	34.8	36.0	40.9	55.9	71.4	75.6	80.6	74.6	63.7	52.1	45.0	39.2
		MCWB	34.1	34.7	36.1	44.3	55.7	59.9	65.5	61.8	56.9	49.7	43.9	37.9
	10%	DB	33.0	34.4	38.2	51.2	66.8	71.6	76.7	71.0	61.1	50.0	43.0	36.8
		MCWB	32.3	33.5	34.6	42.2	53.4	58.5	63.8	60.5	56.3	47.8	41.9	36.0
	0.4%	WB	38.0	37.6	41.7	52.4	63.4	67.4	71.5	70.0	61.7	54.2	48.6	45.7
		MCDB	38.4	39.1	46.2	63.0	72.4	77.4	80.7	79.7	67.4	56.1	50.2	46.9
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	35.9	36.2	38.9	49.1	60.3	64.4	69.3	66.5	60.1	52.0	46.4	40.5
		MCDB	36.6	37.6	43.3	59.3	69.8	74.9	78.8	73.7	64.5	53.3	47.7	41.4
	5%	WB	34.2	35.0	37.1	45.8	57.8	62.1	67.5	64.0	58.7	50.2	44.1	38.1
		MCDB	34.7	36.2	40.1	53.0	67.2	72.2	76.3	70.8	62.5	51.6	44.8	39.0
	10%	WB	32.6	33.6	35.3	43.3	55.1	59.9	65.6	62.4	57.1	47.9	42.2	36.0
		MCDB	33.1	34.4	37.7	49.8	66.9	69.1	74.3	68.7	60.2	49.5	42.9	36.6
(35) Mean Daily Temperature Range	5% DB	MDBR	9.6	9.5	13.3	16.5	19.2	17.1	16.8	15.7	12.8	9.4	6.2	6.8
		MCDBR	6.1	8.2	15.6	25.5	27.4	24.5	22.4	22.0	16.9	11.8	7.9	8.2
		MCWBR	6.0	6.8	10.1	13.5	11.7	9.6	7.8	9.2	9.9	8.7	7.5	7.9
	5% WB	MCDBR	6.5	7.8	13.4	20.9	22.8	20.7	19.6	18.0	14.3	10.2	7.7	7.8
		MCWBR	6.5	6.8	9.3	12.5	11.1	9.3	8.2	8.8	9.1	8.7	7.4	7.6
(40) Clear-Sky Solar Irradiance	taub		0.255	0.297	0.324	0.364	0.361	0.356	0.379	0.366	0.342	0.322	0.279	0.366
	taud		2.087	2.156	2.234	2.340	2.417	2.457	2.409	2.437	2.497	2.408	2.179	2.603
	Ebn at Noon		145	211	250	263	273	277	266	260	247	207	135	97
	Edh at Noon		12	23	29	32	32	32	32	29	23	18	11	8
(44) All-Sky Solar Radiation	RadAvg		74	259	727	1188	1604	1680	1644	1266	743	312	84	39
	RadStd		14	58	92	126	179	118	161	144	92	53	15	5

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (53 neighbors)	+0.92	N/A	N/A	N/A	N/A	N/A	-279	-306	N/A	N/A

Nomenclature: See separate page